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Image-based deep learning automated sorting of date fruit

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ABSTRACT

Deep Convolutional Neural Network (CNN) with a unique structure for combining the feature extraction and classification stages, has been considered to be a state-of-the-art computer vision technique for classification tasks. This study presents a novel and accurate method for discriminating healthy date fruit (cv. Shahani), from defective ones. Furthermore, owing to the use of deep CNN, this method is able to predict the ripening stage of the healthy dates. The proposed CNN model was constructed from VGG-16 architecture which was followed by max-pooling, dropout, batch normalization, and dense layers. This model was trained and tested on an image dataset containing four classes, namely Khalal, Rutab, Tamar, and defective date. This dataset was collected by a smartphone under uncontrolled conditions with respect to illumination and camera parameters such as focus and camera stabilization. The CNN model was able to achieve an overall classification accuracy of 96.98%. The experimental results on the suggested model demonstrated that the CNN model outperforms the traditional classification methods that rely on feature engineering for discrimination of date fruit images.

1. Introduction

Date fruit (*Pheonix dactylifera* L.) is an essential agricultural product in the Middle East and Northern Africa countries which has a tremendous economic role in these regions. According to the Food and Agricultural Organization (FAO), these countries are the largest date producer in the world. Based on FAO reports, Iran production of date fruit was about 1.19 million tons in 2017 and had the second rank between the top 20 countries production of dates (Food and Agriculture Organization, 2017). Due to its vitamins and minerals content, and also high nutritious values, date fruit can be considered as a candidate to use as health-promoting foods (Vayalil, 2012).

Date has four separate stages of the ripening process which are described traditionally by variation in its color, texture, and taste. These stages are known as Kimri, Khalal, Rutab, and Tamar (Pourdarbani et al., 2015). However, dates are harvested and consumed at three final stages of their ripening, namely Khalal, Rutab, and Tamar. The choice of the ripening stage for harvesting and marketing depends on cultivar, solvable tannins content, climatological situations and market demands (Mireei et al., 2010; Awad, 2011). Unfortunately, during the growth and ripening process and harvesting stage, some of the product is damaged by some pests, insects, mites, and mechanical equipment and these defects cause significant economic losses to the storage and exportation of date fruit (Mireei and Sadeghi, 2013).

For two main reasons, dates with different maturity stages are not recommended to be packed in the same package. First they can have destructive mutual effects, and second, in order to improve the marketing conditions, different customer tastes should be considered (Pourdarbani et al., 2015). Also, traditional sorting methods of dates are time-consuming, costly, and tedious. Thus, automatic sorting systems should be developed to conserve energy, improve the quality of packed product and gain customer satisfaction. So non-destructive methods are needed to detect the maturity stages of the date. In this context, some of the researchers have introduced such systems. Muhammad (2015) used different distinguished features containing color, shape, and texture to classify various types of dates. Texture information was extracted from color spaces (RGB and YCbCr color spaces) by Local Binary Pattern (LBP) and Weber Local Descriptor (WLD). Then, Fisher Discrimination Ratio (FDR) algorithm was utilized to choose the best features. In the end, support vector machines (SVMs) were applied as a classifier. Al Ohali (2011) utilized a neural network to classify dates into three categories. Extracted features were size, shape, and intensity. In addition to the machine vision systems, some researchers used near-infrared spectroscopy as a non-destructive method to discriminate between types and quality of the date fruit at various stages of ripening (Mireei et al., 2010; Alhamdan and Atia, 2017).

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Nomenclature

a_i & a_j	Softmax input for class i & j	x	Current batch
C_j	Actual classes	*FP*	*False positives*
C_i^*	Classifier classes	*FN*	*False negatives*
TN	True negatives	*TP*	*True positives*
P_i	Normalized probability value	μ	Mean
		σ	Standard deviation
		γ, β, $\in$	Constant parameters

Despite the successful uses of the techniques based on feature engineering in the classification tasks, these methods are complex because they need distinct sections, i.e., feature extraction, feature selection, and feature learning. So a feature-engineering-based system maybe not optimally perform, and there is a need for new approaches which can handle these distinct sections.

Machine learning and computer systems have been increasingly utilized in the wide range of technologies from searching in the web to features extraction for image classification tasks, face detection in cameras or smartphones, and object detection in an image. In recent years, these applications have used deep learning method due to its capability to perform automated feature extraction process directly from images (LeCun et al., 2015). Due to this ability and also the capacity to learn complicated and huge issues, deep learning has offered excellent efficiency in vision-based tasks such as pattern recognition, classification. (Amara et al., 2017). Development of deep learning methods has led to the integration of two critical steps in image processing, namely the learning of the most appropriate extracted feature and dataset classification.

One particular class of deep learning techniques is convolutional neural network which includes a collection of non-linear transformation functions. CNN has been recently used in the area of agriculture and food production. Many researches have been published since 2015, and the main subject of these articles was object detection and image classification. In this regard, some articles can be mentioned as examples: detect and count the number of fruit such as tomatoes, apple, mangoes, and almonds in the image (Sa et al., 2016; Rahnemoonfar and Sheppard, 2017; Bargoti and Underwood, 2017; Chen et al., 2017), classification of different leaf and leaf disease (Lee et al., 2015; Hall et al., 2015; Mohanty et al., 2016; Sladojevic et al., 2016; Amara et al., 2017), identification weed in agricultural fields for robotic platforms (Potena et al., 2016; Milioto et al., 2017; Dyrmann et al., 2017; McCool et al., 2017). Furthermore, Yu et al. (2018) developed a deep learning approach for predicting firmness and soluble solid content of postharvest Korla fragrant pear using Vis/NIR hyperspectral reflectance imaging.

Unlike the Artificial Neural Networks, CNNs are able to automatically perform both the features extraction and selection by virtue of the network's depth and weight sharing between nodes which can help alleviate over-fitting (Abdel-Hamid et al., 2013). Since CNN learning incorporates millions of parameters, a large set of training data and huge computation are required for the training process. Specific strategies are introduced to compensate for these limitations and have been known as transfer learning. Transfer learning method includes using a pre-trained deep network as a feature extractor and fine-tuning pre-trained deep network weights by a new dataset (Pan and Yang, 2010; Ghazi et al., 2017).

Automatic classification is a crucial parameter in the date fruit

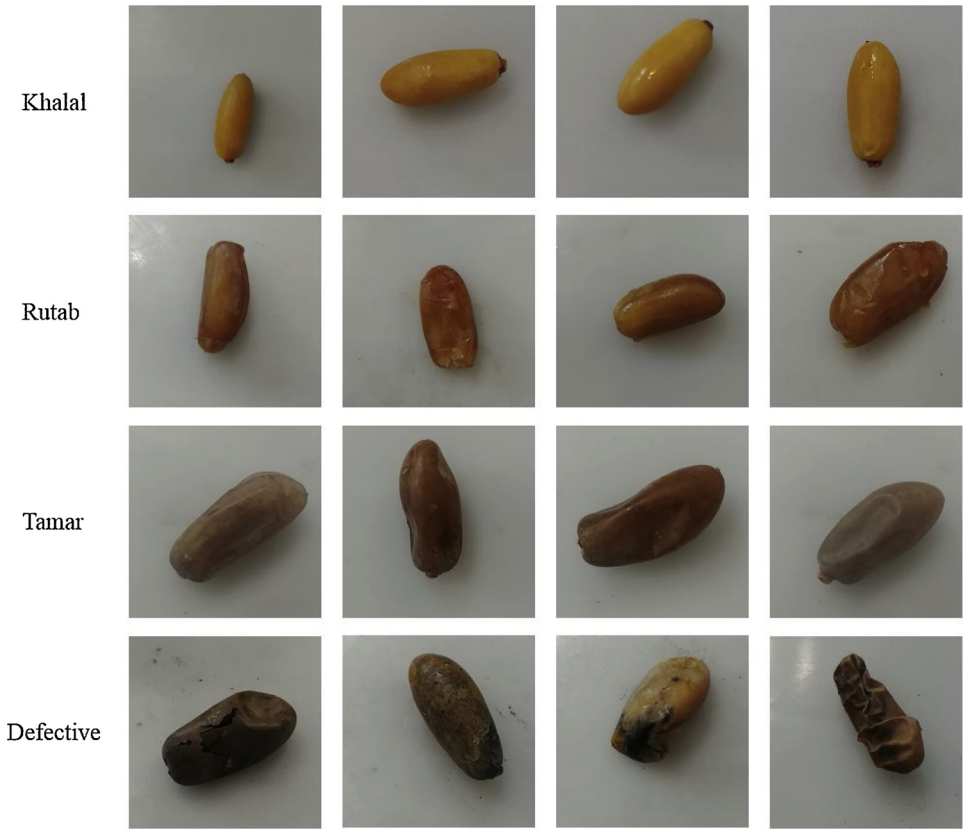


Fig. 1. Example images of Shahani cultivar of date fruit at various maturity stages and defective samples.

industry to increase the speed of the process and maintain the uniformity in date packages. Computer vision-based automated systems used for classification and sorting date fruit utilize binary images, That is, the segmentation of the object from the background is a necessary step, and only once this is achieved can the features of the segmented region be extracted. On the other hand, the main idea of CNNs is to construct a hierarchy of self-learned features. Compared to the traditional classification methods, self-learned features cause the CNN to be less affected by natural changes like illumination variations. This ability removes the need to segment the object from the background and enables the network to learn new date fruit species with little effort. Accordingly, the main contribution of this study is utilizing CNN to construct an automated date sorting system which eliminates the need for segmenting the object from background leading to a system that its performance is independent of the imaging conditions.

2. Materials and methods

2.1. Date fruit samples

In this study, Shahani date cultivar which is considered as wet date fruit is selected. Due to its delicious taste, Shahani is one of the most popular date fruit in Iran. The best kind of Shahani date is cultivated commonly in Jahrom which is one of the most important regions of horticultural products in the south of Iran (Najafi and Khodaparast, 2009; Mireei et al., 2010). Shahani date fruit from every three stages of maturity namely Khalal, Rutab, and Tamar were collected in October 2018 from an orchard located in Jahrom. An LG-V20 smartphone camera was used to capture an image of each sample. The background of each image was kept constant. However, other parameters such as focus, the angle of image capturing, lighting conditions, and camera distance from the samples were not fixed. The collected dataset comprised over 1300 images of healthy and defective dates. Each date image was manually classified based on the different growth stages and defective dates by experts. Therefore, the four classes considered here were Khalal, Rutab, Tamar, and defective date. These classes included 327, 288, 284, and 458 images, respectively. In this study, defective dates such as damaged dates by insects, mites or birds, blemished date fruit, dates that had been mechanically damaged, unripe dates, and molded dates were determined by Codex Standard 143−1985. Fig. 1 shows some of the acquired images of the date fruit samples.

2.2. Image preprocessing and data augmentation process

Too many parameters are embedded in deep convolutional networks. Therefore the training phase requires a large dataset of training images to learn all the parameters, otherwise, the CNN would not be appropriately trained and run the risk of over-fitting. Since it is not often possible to increase the number of input images without a parallel rise in cost and time a strategy known as data augmentation needs to be applied. In data augmentation method the training dataset is increased with the same label or class to prevent the over-fitting issue. This technique is done through image processing methods such as flipping horizontally or vertically the input images (Sladojevic et al., 2016). In the present study, the training dataset was augmented using rotation, height shift, width shift, zoom, horizontal-flip, and shear intensity. The data augmentation technique created 37,056 images for the training process.

2.3. Convolutional neural network architecture

In order to train the input RGB image, CNNs can involve the convolutional layers as one of the major layers. The task of these layers is to extract low-level features such as edges, blobs, and colors by filters (kernels). The convolutional operator is invariant to translation however not rotation invariant (Dyrmann et al., 2016). Besides, other common layers are pooling layers which have the role of reducing the size of images (Farooq and Sazonov, 2017). Moreover, like neural networks, the structure of CNNs consists of fully connected layers. Training CNNs includes two steps: Feed-forward and back-propagation. In the Feed-forward step, the network error is calculated by comparing the difference between the network's output on the basis of the given image to the network as input, and the actual labeled output. In the back-propagation stage, the gradient of parameters are calculated depending on the network error and then each weight matrix is updated based on the calculated gradient (Rahnemoonfar and Sheppard, 2017).

There are various dominant pre-trained structures of CNN which have been successfully trained by a major dataset of labeled images such as ImageNet with 1000 different classes to do classification tasks (Russakovsky et al., 2015). Since the images of such datasets are in sharp contrast to the images of the date fruit, the features extracted by these networks may be not suitable for classifying the various maturity stages of dates and defective samples. To overcome this drawback, a technique for model reuse known as the fine-tuning process is applied. It has become standard to utilize pre-trained CNN models to initialize the network weights and transfer the learned features to a new task (fine-tune), which has fewer categories. In this process, the parameters of the model are updated by the new dataset. That is, the network training process begins with the weights matrix of the existent network rather than starting with a random weights matrix (Sa et al., 2016). Common pre-trained CNN consists of AlexNet (Krizhevsky et al., 2012), GoogLeNet (Szegedy et al., 2015), ResNet (He et al., 2016) and VGG (Simonyan and Zisserman, 2014).

The depth of CNN plays a major role in the classification accuracy and appropriate detection performance so that the classification error decreases with increasing the depth of CNN (Simonyan and Zisserman, 2014). Visual Geometry Group network (VGGNet) as the first runner-up in ILSVRC-2014, utilizes a homogeneous structure to employ the beneficial effects of increased CNN depth on performance (Ghazi et al., 2017). The very depth VGGNet significantly outperforms the

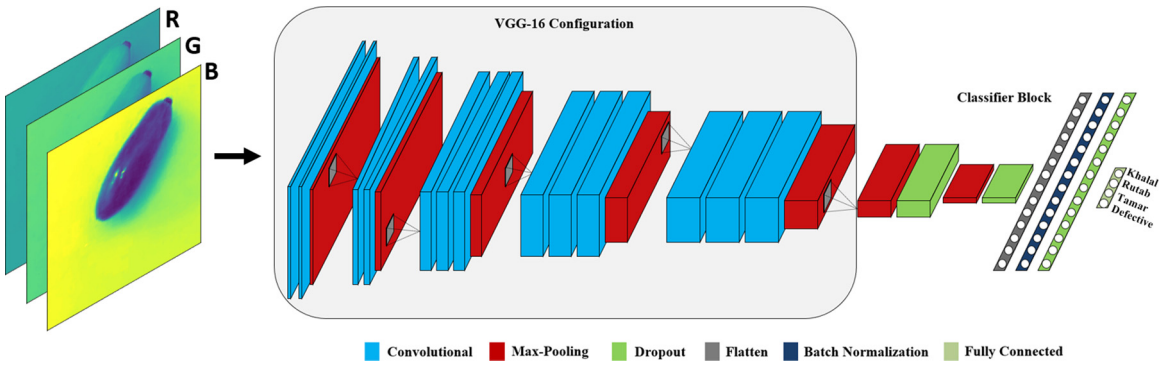


Fig. 2. Network architecture including VGG-16 structure and modified classifier block.

architectures which achieved the best results in the ILSVRC-2012 (AlexNet) and ILSVRC-2013 (ZFNet (Zeiler and Fergus, 2014)). Despite the VGGNet classification power is slightly inferior to GoogLeNet (the winner of ILSVRC-2014), its network topology is less complex than GoogLeNet and features generated by VGGNet architecture outperform other CNNs, such as AlexNet and GoogLeNet (Simonyan and Zisserman, 2014; Sa et al., 2016). Thus it is one of the popular choices in machine learning and computer vision tasks. As above mentioned and because of VGGNet's proper performance in the classification tasks (Sa et al., 2016; Ghazi et al., 2017), VGGNet was selected for the present study. VGGNet consists of five various blocks which are set homogeneously and sequentially so that the output of each block is defined as the input of the next block (Fig. 2). By this architecture, the network extracts powerful features from the input images such as texture, shape, and color.

VGG-16 is one of the two VGG architectures introduced in Simonyan and Zisserman (2014) which contains 13 convolutional layers with 3×3 kernels and five 2×2 max-pooling layers. Max-pooling is a subsampling layer that reduces the dimensional of the feature map by keeping the maximum value of sliding windows within each features map. Features map is the output of each convolution or pooling layer. The activation function for each convolutional layers is the ReLU (Rectified Linear Unit) function (Simonyan and Zisserman, 2014; Ghazi et al., 2017). This function performs the following mathematical operation on each input data (Dyrmann et al., 2016):

$$f(x) = \begin{cases} x, & if\ x > 0 \\ 0, & otherwise \end{cases} \quad (1)$$

For maturity stages diagnosis of the healthy dates and separation of the defective dates from healthy samples, a modified model was developed to classify four types of date conditions namely Khalal, Rutab, Tamar, and defective date (Fig. 2). The VGG-16 configuration-based modified model was followed by a classifier block. The classifier block contained two max-pooling layers with a 2×2 window. After each max-pooling layer, dropout layers were added for regularization. Then flatten layer was applied to flatten the output of the second dropout layer. The output of this layer was connected to the batch normalization layer, and the batch normalization output was passed to the third dropout layer. Finally, the output of the last dropout layer was crossed through a fully connected layer having four neurons, each corresponding to the probability of four classes of the date conditions. The CNN configuration, developed in this research, is outlined in Table 1. In this table, layers output shape are denoted as (height × width × number of channels).

2.3.1. Dropout layer

Dropout is a regularization method that alleviates the overfitting risk of training CNN. At any stage of the training process, it randomly removes some neurons to allow the remaining neurons to participate in the process of network training and their weights to get updated. This process also helps the neurons to learn more features (Srivastava et al., 2014).

2.3.2. Batch normalization layer

After any transformation process of data in deep convolutional neural networks, normalization of data is a great issue because it changes the distribution of layer's input. The use of batch normalization layer in CNNs causes the network to become deeper and the number of required iterations for the training process to decrease. This method holds the inputs of layers on the same range of values with the following equation (Ioffe and Szegedy, 2015; Dyrmann et al., 2016):

$$y = \frac{x - \mu}{\sqrt{\sigma^2 + \in}} y + \beta \quad (2)$$

where x is the current batch, μ and σ are the mean and standard deviation of the x, respectively, γ and β stand for trainable parameters and a small constant ε is applied to avoid zero-division.

2.3.3. Fully connected layer

In the fully connected layer, also known as the dense layer, all input neurons are connected to all neurons of output, as is typical in traditional artificial neural networks. So, this layer causes the spatial information to be removed. These layers are utilized to map spatial learned features to image labels (Dyrmann et al., 2016).

2.4. Fine-tuning the network

The modified network utilized the RGB input images with the labeled data. For the network training process, 30,688 images for training and 6,368 for validation were utilized. Also, 199 images were used for the testing process. During the training process, the dropout rate was set to 50% for all dropout layers. A softmax activation function was applied to the fully connected layer, which predicted the normalized probabilities of a test image belonging to each of the date classes. Softmax is the common and standard function utilized in the classification tasks. A dense layer with softmax function has n neurons for n classes task. If p_i is the normalized probability value of any neuron, given by Eq. (3), therefore, $\sum_{i=1}^{n} p_i = 1$.

$$p_i = \frac{\exp(a_i)}{\sum_{j=1}^{n} \exp(a_j)} \quad (3)$$

where a_i is the softmax input for node i (class i), and $i,\ j \in \{1,\ 2,\ ...,n\}$ (Tang, 2013).

Fine-tuning was carried out based on four classes of the date conditions, namely Khalal, Rutab, Tamar, and defective date; thus, the dense layer had four neurons. In Fine-tuning process, first, the modified CNN weights were pre-trained for 10 epochs on the IamgeNet dataset. In this step, all convolutional based blocks of the model (VGG-16 network) were frozen then the classifier block was trained by random weights. The classifier block was already trained to prevent the error signal propagation through the network during the training process. In order to obtain the best result and improve the classification accuracy, the modified model was trained five times on training dataset. As it can be observed in Fig. 2, in the first step, only the last convolutional block was unfrozen, and then this block with the classifier block of the

Table 1
The developed CNN configuration.

Input (150 × 150 RGB Image)								
Conv Block	Conv		Conv		Conv		MaxPooling	
1	150 × 150 × 64		150 × 150 × 64		–		75 × 75 × 64	
2	75 × 75 × 128		75 × 75 × 128		–		37 × 37 × 128	
3	37 × 37 × 256		37 × 37 × 256		37 × 37 × 256		18 × 18 × 256	
4	18 × 18 × 512		18 × 18 × 512		18 × 18 × 512		9 × 9×512	
5	9 × 9×512		9 × 9×512		9 × 9×512		4 × 4×512	
Classifier Block	MaxPooling	Dropout	MaxPooling	Dropout	Flatten	Batch Normalization	Dropout	Fully Connected
	2 × 2×512		1 × 1×512		512		512	4

network were trained. In the second step, Conv-blocks 4 and 5 were unfrozen, and the CNN model was trained anew. This process continued until all convolutional blocks were unfrozen and along with the classifier block were trained again.

In this research, Python 3.6 using Keras library with Tensorflow backend were applied to carry out all image processing steps, build and train the modified convolutional neural network. A mini-batch of 32 images, RMSProp optimizer (Tieleman and Hinton, 2017), and the cross-entropy loss function (Drozdzal et al., 2016) were used to train the modified CNN on date images.

2.5. Performance evaluation

Confusion matrix was extracted to evaluate the classification performance. The data in confusion matrix expresses the actual class labels and the labels predicted by the CNN classifier. The basic structure of the confusion matrix for a multi-class (k-class) classification is presented in Fig. 3, which shows how the samples are distributed over actual (columns) and predicted (rows) classes.

The phrase (n_{ij}) expresses the samples that are placed in class number j using the (i.e. C*j) classifier, but in fact, they belong to the class number i (i.e. C_i). Therefore, the elements on the main diameter (i = j) represent the correct classification, while off-diagonal elements (i ≠ j) indicate incorrect classification of samples. Consideringclass i, there can be four type of cases: True Positives (TP) or samples that are correctly classified in class i. False Positive (FP) or samples that are incorrectly classified in class i. True Negative (TN) or samples that are correcltly rejected from class i. False Negative (FN) or samples that are incorrectly rejected from class i. The corresponding counts are specified as $n_{TP} = n_{i,i}$, $n_{FP} = n_{+,j} - n_{i,i}$, $n_{FN} = n_{i,+} - n_{i,i}$ and $n_{TN} = n - n_{TP} - n_{FP} - n_{FN}$ where $n_{i,+}$ and $n_{+,j}$ indicate the total elements on row i and column j in a confusion matrix, respectively (Labatut and Cherifi, 2012).

The performance of the classifier was evaluated based on some statistical parameters of the confusion matrix, such as accuracy, precision, specificity, sensitivity, and area under the curve (AUC). The mentioned measures and their formulas have been presented in Table 2 (Taheri-Garavand et al., 2015).

3. Results and discussions

Image acquisition from healthy and defective dates was carried out by LG-V20 cell phone camera at various capturing conditions for training and testing of the modified CNN. The created dataset contained three different growth stages of dates (Khalal, Rutab, and Tamar) and various dates damaged or decayed due to multiple factors such as insects or birds. Each of the RGB images included only one date picture with identical background.

The modified convolutional neural network presented in Section (2 and 3) was trained by 30,688 and 6,368 images for training and validation, respectively. The introduced model structure used in the present research was a modified version of the VGG-16 convolutional neural network. The VGG16 architecture was modified by adding classifier block instead of the fully connected classifier. the modified classifier block included max-pooling, dropout, batch normalization, and dense layers.

The modified CNN model was trained five times on training dataset, and 25 epochs were used for each training time. In each training step, one of the convolutional blocks was brought out of the freeze mode. Final classification performance of each training time is reported in Table 3. According to this table, the best performance was achieved when all convolutional blocks of modified CNN were trainable. For this network, the accuracy and loss of training and validation data for each epoch are presented in Fig. 4. The training process was stopped after epoch 15 to achieve the highest accuracy of classification without any over-fitting. As it can be seen from Fig. 4, after this epoch, the curves of validation accuracy and loss start to smooth and the gap between curves of both accuracy and loss regarding the validation and training data increases. At the 15th epoch, the classification accuracy of the training and validation dataset were 0.9794 and 0.9846, and their cross-entropy loss values were 0.0602 and 0.0522, respectively. Also, prediction accuracy and cross-entropy loss value of the test data were obtained 0.9698 and 0.0742, respectively.

3.1. Performance evaluation of the CNN model

The filters or features extracted by the model from various layers specify the final accuracy of the modified model. A convenient way to appraise the performance of the modified model to extract low-and high-level features is the evaluation of the visual patterns of the filters' response. For this aim, various filters were extracted from different convolutional layers of the model. Fig. 5 shows some filters used in the first, middle, and last convolutional layers. As it can be seen from this figure, colorful features and directional edges were encoded by the first layer filters. The filters from middle convolutional layer encoded simple textures which were made from compositions of colors and edges. Filters of the final convolutional layer extracted the textures and specific patterns of the images. Thus, different efficient filters have been simulated by the modified model based on the shape, color, and texture.

Fig. 6 illustrates the visualization of the activations (feature map) in the first convolution layer of the modified CNN model (Fig. 2), for a given image. The feature map reveals how the convolutional processing decomposes the input image into several filters. Also, the Gradient-weighted Class Activation Mapping (Grad-CAM) technique which is helpful in debuging the prediction process, was applied to visualize the used areas of random input images to extract feature for prediction of the class of the images. In this method, the class activation heatmap of the input image is produced and calculated for all of its locations. The 2D grid class activation heatmap determines the importance grade and similarity level of any image locations with respect to the specific class. In order to obtain class activation heatmap, first, the feature map of the last convolution layer for a given image was extracted, and the gradient of the specific class was computed with regard to the feature map of the convolutional layer. Then these gradients were used to calculate the neuron weights. Finally, any channel of the feature map was weighted by these weights. Fig. 7 shows the result of the implementation of this method on random images.

According to Figs. 6 and 7, the CNN model could correctly extract features from the correct location even when a small part of the date fruit was separated due to damage (defective date, see Fig. 7) which demonstrates the capability and stability of the modified model. Thus

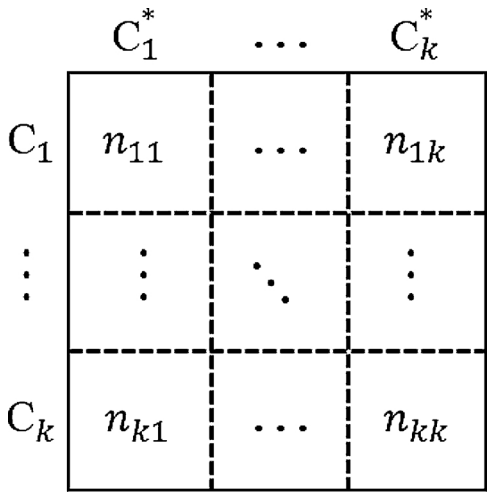


Fig. 3. Confusion matrix for k classes' classification.

Table 2
Statistics parameters of confusion matrix and related formulas [50].

Measure	Formula	Evaluation focus
Accuracy	$\frac{n_{TP}+n_{TN}}{n_{TP}+n_{TN}+n_{FP}+n_{FN}}$	Overall effectiveness of the classifier
Precision	$\frac{n_{TP}}{n_{TP}+n_{FP}}$	Class agreement of the data labels with the positive labels defined by the classifier
Sensitivity(Recall)	$\frac{n_{TP}}{n_{TP}+n_{FN}}$	A measure of the ability of classifier to select instances of a certain class
Specificity	$\frac{n_{TN}}{n_{TN}+n_{FP}}$	How effectively a classifier identifies negative labels
AUC	$\frac{1}{2}\left(\frac{n_{TP}}{n_{TP}+n_{FN}}+\frac{n_{TN}}{n_{TN}+n_{FP}}\right)$	The classifier's ability of classifier to avoid false classification

Table 3
Comparison of the modified CNN performance in different condition of convolutional blocks.

Unfrozen block	Training		Validation	
	Accuracy	Loss	Accuracy	Loss
5	0.8845	0.3236	0.8571	0.4136
4, 5	0.9202	0.2340	0.8643	0.4012
3, 4, 5	0.9370	0.2155	0.8714	0.3873
2, 3, 4, 5	0.9517	0.1697	0.8643	0.3789
1, 2, 3, 4, 5	**0.9794**	**0.0602**	**0.9846**	**0.0522**

the modified CNN model can be utilized to discern the defective and healthy dates and predict the various growth stages of healthy date.

3.2. The confusion matrix results

In order to evaluate the performance of the classifier in identification of date conditions, the confusion matrix and the performance parameters of confusion matrix are presented in Fig. 6 and Table 4, respectively. The Confusion matrices in Fig. 8 show the distribution of predictions for each of the four date conditions. Here it can be seen that Rutab and Tamar were correctly classified with accuracy of 100%. Khalal and defective dates had three misclassified images which confused with dates in the Rutab and Tamar class, respectively. However, all incorrectly-classified images were distributed next to the main diagonal of the confusion matrix, which means that they were predicted close to the correct classes.

As shown in Table 4, the average amount of the performance parameters of the classifier including accuracy, precision, sensitivity, specificity and area under the curve (AUC) were obtained 98.49, 96.63, 97.33, 99.05, and 98.19, respectively. Also, the amount of accuracy for different classes; Khalal, Rutab, Tamar and defective date was calculated as 98.49. It can be seen that the Rutab and Tamar classes were classified with the AUC values of 99.04 and 99.06 percent, respectively. This which shows the excellent accuracy of Rutab and Tamar classification compared with other classes.

In the following, results of other researchers who have worked on the prediction of growth stages of date using computer vision have been presented. Pourdarbani et al. (2015) developed a machine vision-based sorting system to sort date fruit based on various growth stages of date namely Khalal, Rutab, and Tamar. They used color features extracted from 300 samples to determine the maturity stages of date. The classification accuracy was obtained 88.33%. Muhammad (2015) extracted various features containing color, shape, and texture from 800 date fruit to classify different types of dates (Ajwah, Sagai, Sellaj, and Sukkary). Fisher discrimination ratio (FDR) algorithm and support vector machines (SVMs) were utilized as a features selector and classifier, respectively. The suggested method achieved 98.1% accuracy. Al Ohali (2011) used size, shape, and intensity features extracted from 660 images to classify the date fruit. In this research, the back-propagation neural network was used as a classifier. The classification accuracy of dates based on three levels of quality was calculated as 80%. Lee et al. (2008) applied color features for automated distinguishing of dates according to seven maturity levels. The developed technique was tested on 700 images, and the classification accuracy was obtained 90.86%.

Controlling the conditions of image acquisition, features extraction, adequate feature selection, and optimal classifier technique are essential in the feature-engineering-based methods for appropriate results. In contrast to these methods, the images used in the present research were shot under uncontrolled conditions with regard to lighting conditions and camera stabilization and parameter. Also, the feature extraction and classification steps were combined by the CNN model. Thus the proposed method in this study exhibits not only less complexity to be performed but also a satisfactory accuracy of 96.98% (See Table 4). These results are due to the capability of the CNN models in the classification problems.

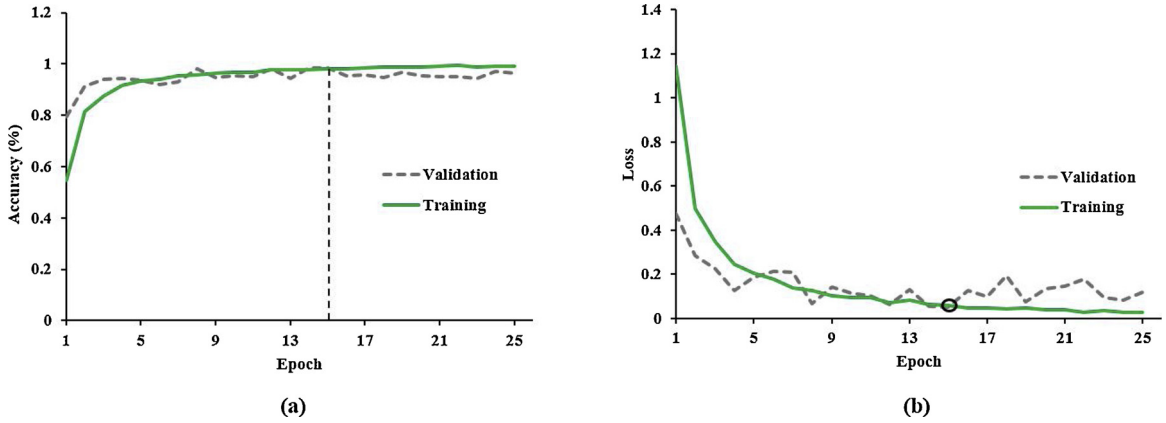


Fig. 4. Classification accuracy (a) and cross-entropy loss (b). The marked points at epoch 15 show where the highest accuracy of classifier task was achieved.

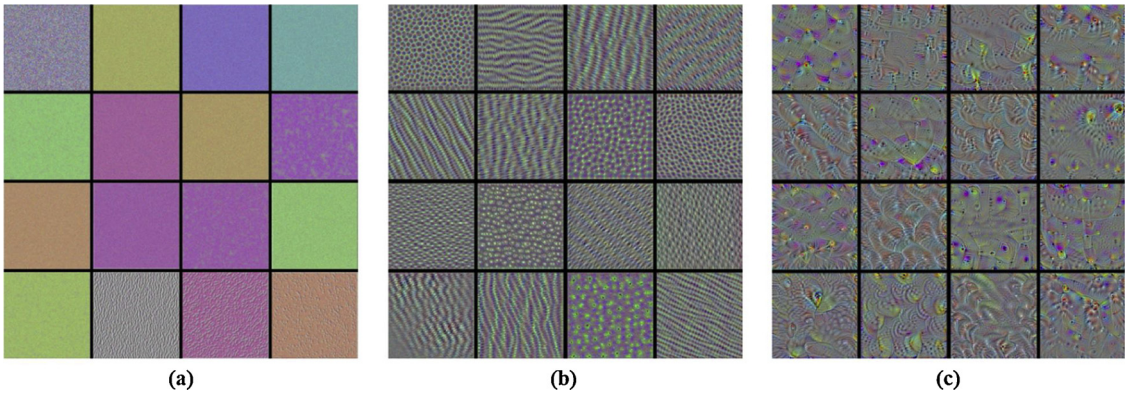


Fig. 5. The CNN model filter banks of the first (a), the middle (b), and the final convolutional layers (c).

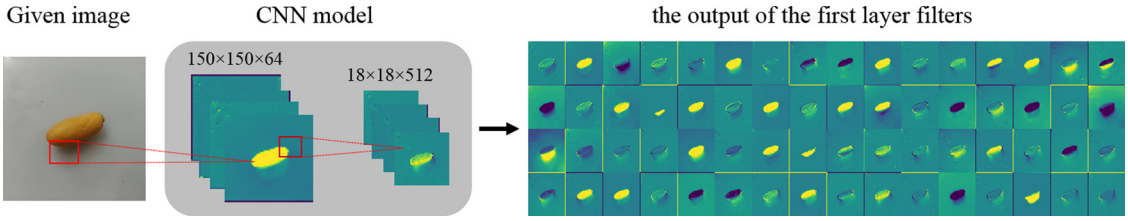


Fig. 6. Every channel of the first activation layer on the test image.



Fig. 7. a) Orginal images for classifier, b) Grad-CAM visualization.

Table 4
The performance results of modified CNN classifier.

Class	AC (%)	PR (%)	SE (%)	SP (%)	AUC (%)
Khalal	98.49	100	93.75	100	96.88
Rutab	98.49	93.48	100	98.08	99.04
Tamar	98.49	93.02	100	98.11	99.06
Defective	98.49	100	95.59	100	97.79
Average per class	98.49	96.63	97.33	99.05	98.19

4. Conclusions

Classifying date fruit based on the various maturity stages is a main process in the food industry. The traditional classifier methods consist of several separate steps. In this study, a deep CNN was constructed to overcome the complexity of the traditional systems, distinguish the defective dates from healthy ones, and predict the ripening stage of the healthy dates (cv. Shahani).

The introduced CNN model in present research was constructed from VGG-16 architecture and classifier block. This block contained max-pooling, dropout, batch normalization, and dense layers. The CNN model was utilized to classify date fruit into the four classes namely, Khalal, Rutab, Tamar, and defective dates. All photographs were taken via LG-V20 smartphone under various conditions of image acquisition. The study seems to be state-of-the-art in terms of uncontrolled conditions. The average per class of classification accuracy, precision, specificity, sensitivity, and AUC, which describe the performance of the modified model, ranged from 96% up to 99%. These results showed the high classification performance of the CNN model to detect healthy from defective dates and predict the maturity stage. Therefore, the deep CNN models can be employed in modern food industries and smartphone-based applications used by consumers.

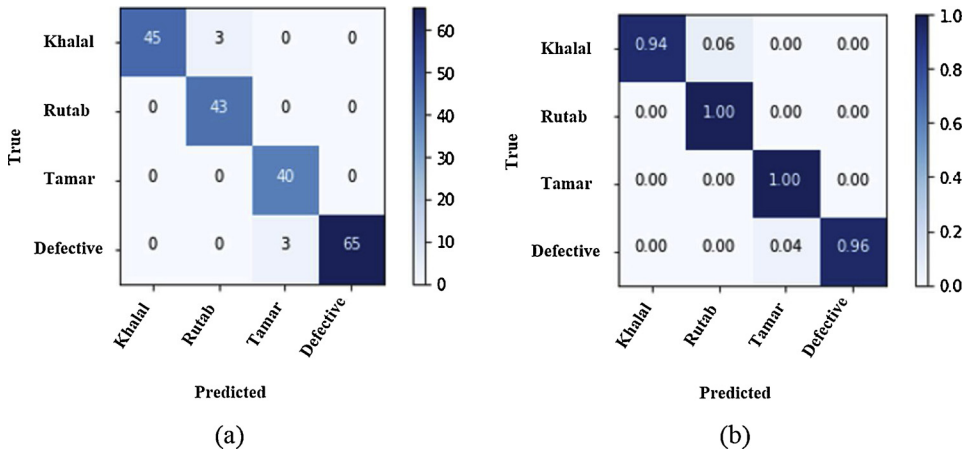


Fig. 8. Distribution of predicted the four classes of date conditions: (a) confusion matrix, (b) normalized confusion matrix.

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